

Lab 9: RISC Advanced I/O

This lab exercise uses the Logisim file mips-lab.cir provided in Lab-7. You can use a machine code assembler tool like Mars ([download link](#)) to quickly compile the machine code.

1. Write a program named Prog1.txt to read the switch field from the input register and display it directly on the LED bank.
2. Write a program named Prog2.txt that reads the state of button B0. When button B0 is pressed, it toggles the state of LED L0.
3. Use the three buttons as commands. When B0 is pressed, show the raw switch value. When B1 is pressed, show the switch value shifted left by one bit. When B2 is pressed, show the bitwise inversion of the 10-bit switch field. The program is named Prog3.txt.
4. Create a counter stored in RAM named Prog4.txt . Use B0 to increment the counter, B1 to decrement it, and B2 to reset it to zero. Display the current counter value on the LED as binary value.
5. Add a two-digit 7-segment display block to the circuit. Write a program named Prog5.txt to display the value of a counter. The counter increments when button B1 is pressed and decrements when button B0 is pressed. Pressing button B2 resets the counter to 0.
6. Add an 8×8 LED matrix block to the circuit. Write a program named Prog6.txt such that when button B1 is pressed, the displayed symbol on the LED matrix changes. The symbols should be designed by you and must include a left arrow, right arrow, upward arrow, and downward arrow.